



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 26: Externalities II

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ICS2523 Microeconomics

Lecture 26

Last lecture, we started studying **Externalities**

- Definition
- Allocative inefficiency

In today's lecture, we will finish with externalities:

- Solutions to externalities problem

And then, we will move to another market failure:

- **Public Goods**

Homework 1 Lecture 25

Suppose only two firms produce a certain medical device used in surgery. The medical device is produced at a constant average and marginal cost of \$10, and the demand for the device is given by

$$Q = 5000 - 100P$$

- a) If a **Cournot game** is played in a single period, find each firm's equilibrium quantities, price and profits.
- b) If there was a monopoly, how many devices would be sold, at what price and how much will the profits be?
- c) Find the condition for the discount factor (δ) so that collusion is attractive in an infinitely repeated game. In this case, they will use the monopoly price in the stage game, and each firm will earn an equal share of the monopoly profit. When one of the firms deviates, the firms play the Cournot Nash equilibrium

Homework 1 L25: Solution

MC = \$10, and the demand for the device is given by: $Q = 5000 - 100P$

Thus, the inverse demand function is: $P = 50 - Q/100$

- a) If a **Cournot game** is played in a single period, find each firm's equilibrium quantities, price and profits.

Each firm maximises profits, for Firm 1:

$$\text{Max } \pi_1(q_1, q_2) = P \cdot q_1 - 10 \cdot q_1 = (50 - (q_1 + q_2)/100) \cdot q_1 - 10 \cdot q_1$$

First order condition:

$$\frac{\partial \pi_1}{\partial q_1} = 50 - q_1/100 - q_2/100 - q_1/100 - 10 = 0$$

$$40 - q_1/50 - q_2/100 = 0$$

Firm 1 Best-response:

$$BR_1(q_2) = q_1 = 2000 - q_2/2$$

Similarly, for Firm 2:

$$BR_2(q_1) = q_2 = 2000 - q_1/2$$

Homework 1 L25: Solution

Thus, the Nash equilibrium is found in the intersection of these two equations:

$$\begin{aligned}q_1 &= 2000 - q_2/2 \\q_2 &= 2000 - q_1/2\end{aligned}$$

$$\begin{aligned}q_1 - 2q_2 &= 2000 - 4000 + q_1 - q_2/2 \\3q_2/2 &= 2000 \\q_{2c} &= 4000/3 \\q_{1c} &= 4000/3\end{aligned}$$

$$\begin{aligned}\text{Price is } P &= 50 - \frac{q}{100} = 50 - \frac{80}{3} = \frac{70}{3} \\ \pi_1 = \pi_2 &= P \cdot \frac{4000}{3} - 10 \cdot \frac{4000}{3} = \frac{160000}{9}\end{aligned}$$

Homework 1 L25: Solution

- b) If there was a monopoly, how many devices would be sold, at what price and how much will the profits be?

In a monopoly:

$$\text{Max } \pi(Q) = P \cdot Q - 10 \cdot Q = (50 - Q/100) \cdot Q - 10 \cdot Q$$

$$\frac{\partial \pi}{\partial Q} = 50 - \frac{Q}{100} - \frac{Q}{100} - 10 = 0$$

(equivalent to MR=MC)

$$\frac{Q}{50} = 40$$
$$Q_m = 2000$$

$$P_m = 50 - \frac{Q_m}{100} = 30$$
$$\pi_m = P \cdot Q - 10 \cdot Q = 60000 - 20000 = 40000$$

Homework 1 L25: Solution

- c) Find the condition for the discount factor (δ) so that collusion is attractive in an infinitely repeated game. In this case, they will use the monopoly price in the stage game, and each firm will earn an equal share of the monopoly profit. When one of the firms deviates, the firms play the Cournot Nash equilibrium.

If the firms collude, they will use the monopoly price and behave “as” a monopoly (although the correct thing is to calculate the cartel option, students were asked to use information in (b) so they do not have to calculate so much). Profits of the monopoly should be split in two:

$$V_{collude} = 20000 + \delta \cdot 20000 + \delta^2 20000 + \dots = \frac{20000}{1 - \delta}$$

Homework 1 L25: Solution

When one of the firms deviates (firm 1), in that period the other firm (firm 2) will comply with the agreement ($q_2 = \frac{Q_m}{2} = 1000$). Firm 1 best-response will be:

$$BR_1(q_2) = q_1 = 2000 - \frac{q_2}{2} = 2000 - 500 = 1500$$

$$\text{Thus, } P = 50 - \frac{q_1 + q_2}{100} = 50 - \frac{2500}{100} = 25$$

For that period, firm 1's profits will be:

$$\pi_1 = P \cdot q_1 - 10 \cdot q_1 = 15 \cdot 1500 = 22500$$

Homework 1 L25: Solution

After that period, we assume that the firms use the grim strategy of reverting to Cournot:

$$\begin{aligned} V_{deviates} &= 22500 + \delta \cdot \frac{160000}{9} + \delta^2 \cdot \frac{160000}{9} + \dots \\ &= 22500 + \frac{160000/9 \cdot \delta}{1 - \delta} \end{aligned}$$

To find the discount factor that makes collusion attractive:

$$\begin{aligned} V_{collude} &\geq V_{deviates} \\ \frac{20000}{1 - \delta} &\geq 22500 + \frac{160000/9 \cdot \delta}{1 - \delta} \end{aligned}$$

$$20000 \geq 22500 \cdot (1 - \delta) + \frac{160000}{9} \cdot \delta$$

$$\begin{aligned} \delta \cdot \frac{42500}{9} &\geq 2500 \\ \delta &\geq \frac{2500 \cdot 9}{42500} = \frac{9}{17} = 0.5294 \end{aligned}$$

Homework 2 Lecture 25

Suppose two newsprint producers are located along a river. The upstream firm has a production function of the form:

$$x = f(l_x) = 2000\sqrt{l_x}$$

Where l_x is the number of workers hired per day and x is newsprint output in feet. The downstream firm has a similar production function, but its output may be affected by the chemicals the upstream firm dumps into the river as it produces more x :

$$y = f(l_y, x) = 2000\sqrt{l_y} \cdot (1 + \alpha x)$$

If $\alpha = 0$, then the upstream firm's production has no effect on the downstream firm, but if $\alpha < 0$, then an increase in x harms the downstream firm's production. The newsprint sells for $P = \$1$ per foot and workers earn $w = \$100$ per day.

- If $\alpha = 0$, find how many workers each firm hires and how much newsprint will be produced by each firm.
- If $\alpha = -1/40000$, find how many workers each firm hires and how much newsprint will be produced by each firm.
- If both firms merge, find how many workers each firm hires and how much newsprint will be produced by each firm

Homework L25: Solution

- a) If $\alpha = 0$, find how many workers each firm hires and how much newsprint will be produced by each firm.

$$x = f(l_x) = 2000\sqrt{l_x}$$
$$y = f(l_y) = 2000\sqrt{l_y}$$

The newsprint sells for $P = \$1$ per foot and workers earn $w = \$100$ per day.

If $\alpha = 0 \rightarrow$ then the upstream firm has no effect on the downstream firm, each firm maximizes profits. Each firm will maximize profits by setting

$w = \text{labour's marginal revenue product}$

Homework L25: Solution

- a) If $\alpha = 0$, find how many workers each firm hires and how much newsprint will be produced by each firm.

So, each firm will maximize profits:

$$\pi_x = P \cdot x - TC(x)$$

$$\pi_x = P \cdot x - wl_x = P \cdot x - w \cdot \frac{x^2}{(2000)^2}$$

$$\frac{\partial \pi_x}{\partial x} = P - w \cdot \frac{2x}{(2000)^2} = 0$$

$$x = 2P \cdot \frac{(1000)^2}{w} = 20000$$

$$l(x) = \frac{x^2}{(2000)^2} = 100 \text{ workers}$$

Similarly, $y = 20,000$ feet of newsprint

Homework L25: Solution

- b) If $\alpha = -1/40000$, find how many workers each firm hires and how much newsprint will be produced by each firm.

Firm x's profit-maximizing decision is not affected: hires 100 workers and produces $x=20,000$.

But the downstream firm's marginal product of labour now is lower because of this externality:

$$\begin{aligned}\pi_y &= P \cdot y - TC(y) \\ \pi_y &= P \cdot y - wl_y = P \cdot y - w \cdot \frac{y^2}{(2000)^2 \cdot (1 + \alpha x)^2} \\ &= P \cdot y - w \cdot \frac{y^2}{(2000)^2 \cdot (-0.5)^2} \\ \frac{\partial \pi_y}{\partial y} &= P - w \cdot \frac{2y}{(200)^2 25} = 0\end{aligned}$$

$$y = 2P \cdot \frac{(100)^2 25}{w} = 5000 \text{ feet of newsprint}$$

$$l_y = \frac{y^2}{(2000)^2 \cdot (1 + \alpha x)^2} = 25 \text{ workers}$$

Homework L25: Solution

- c) If both firms merge, find how many workers each firm hires and how much newsprint will be produced by each firm

If both firms merge:

$$\begin{aligned}\pi &= P \cdot (x + y) - TC(x) - TC(y) \\ \pi &= P \cdot (x + y) - wl_x - wl_y \\ &= P \cdot (x + y) - w \cdot \frac{x^2}{(2000)^2} - w \cdot \frac{y^2}{(2000)^2 \cdot (1 + \alpha x)^2} \\ \frac{\partial \pi_x}{\partial x} &= P - w \cdot \frac{2x}{(2000)^2} + 2 \cdot w \cdot \frac{y^2}{(2000)^2 \cdot (1 + \alpha x)^3} \cdot \alpha = 0 \quad (*)\end{aligned}$$

$$\frac{\partial \pi_y}{\partial y} = P - w \cdot \frac{2y}{(2000)^2 \cdot (1 + \alpha x)^2} = 0$$

$$y = P \cdot \frac{(2000)^2 \cdot (1 + \alpha x)^2}{2w}, \text{ replacing in } (*)$$

Homework L25: Solution

- c) If both firms merge, find how many workers each firm hires and how much newsprint will be produced by each firm

$$\frac{\partial \pi_x}{\partial x} = P - w \cdot \frac{2x}{(2000)^2} + 2 \cdot w \cdot \frac{y^2}{(2000)^2 \cdot (1 + \alpha x)^3} \cdot \alpha = 0 (*)$$

$$\frac{\partial \pi_x}{\partial x} = P - w \cdot \frac{2x}{(2000)^2} + 2w \cdot \frac{\left(P \cdot \frac{(2000)^2 \cdot (1 + \alpha x)^2}{2w}\right)^2}{(2000)^2 \cdot (1 + \alpha x)^3} \cdot \alpha = 0$$

$$P - \frac{2w}{(2000)^2} \cdot \left[x - \left(P \cdot \frac{(2000)^2 \cdot (1 + \alpha x)^2}{2w}\right)^2 \frac{\alpha}{(1 + \alpha x)^3}\right] = 0$$

Replacing the values of P , w , and α

$$\left[x + 10000 - \frac{x}{4}\right] = 20000 \rightarrow x = \frac{40000}{3} \rightarrow l_x = \frac{400}{9}$$

$$y = \frac{40000}{3} \rightarrow l_y = \frac{600}{9} = \frac{200}{3}$$

Solutions to the externalities problem

Market Failures

17. Solutions to the externalities problem

1. Merge
2. Government provision
3. Subsidies / grants (positive externalities)
4. Taxation
5. Emissions standards and Fees
6. Pollution rights (Coase Theorem)

17. Solutions to the externalities problem

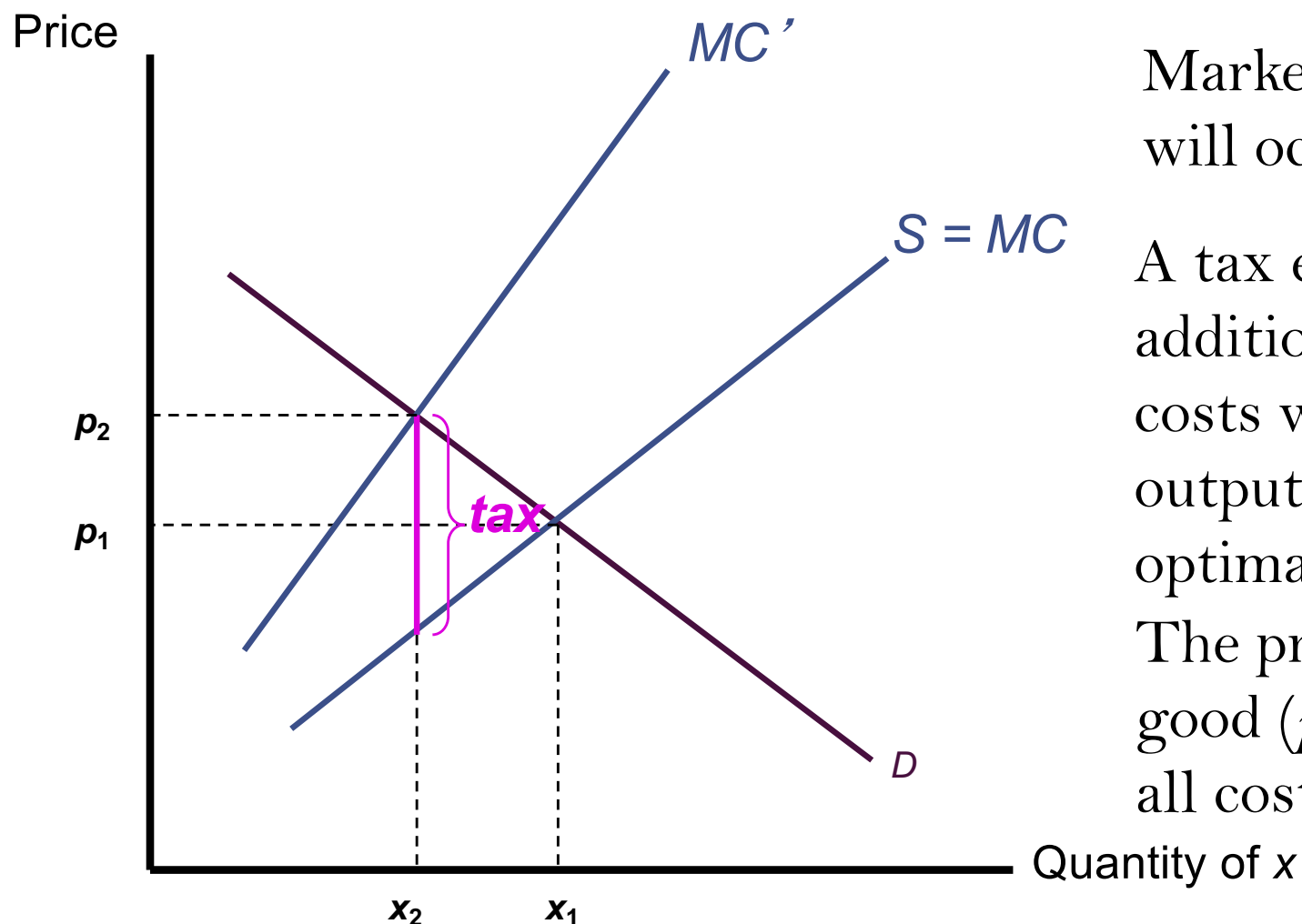
1. Merge
2. Government provision
3. Subsidies / grants (positive externalities)

4. Taxation

5. Emissions standards and Fees

6. Pollution rights (Coase Theorem)

17. Solutions to the externalities problem: **Taxation**



Market equilibrium will occur at p_1, x_1

A tax equal to these additional marginal costs will reduce output to the socially optimal level (x_2)

The price paid for the good (p_2) now reflects all costs

17. Solutions to the externalities problem: **Taxation**

Taxation $\rightarrow$ the optimal Pigovian tax is set to

$$t = -P_y \partial g / \partial x_o$$

(Remember that production of y depends on how much x is used in its production (x_i) and the total quantity of x produced (x_o)) $\rightarrow y_o = g(x_i, x_o)$,

Per unit tax on good x should reflect the **marginal harm** that x does in reducing y output, valued at the market price of good y .

With the optimal tax, firm x now faces a net price of $(p_x - t)$ and will choose y input according to

$$p_y = (p_x - t) \partial f / \partial y$$

17. Solutions to the externalities problem: **Taxation**

The resulting allocation of resources will achieve

$$MRS = p_x/p_y = (1/\partial f/\partial y) + t/p_y = (1/\partial f/\partial y) - \partial g/\partial x_o$$

Problem → depends on the **output level** (should be different for different output levels)

→ regulators must know firm y's production function

17. Solutions to the externality problem:

Emissions Standards and Fees

Emissions Standard

- Establish legal limit to the level/quantity of emissions each firm may produce

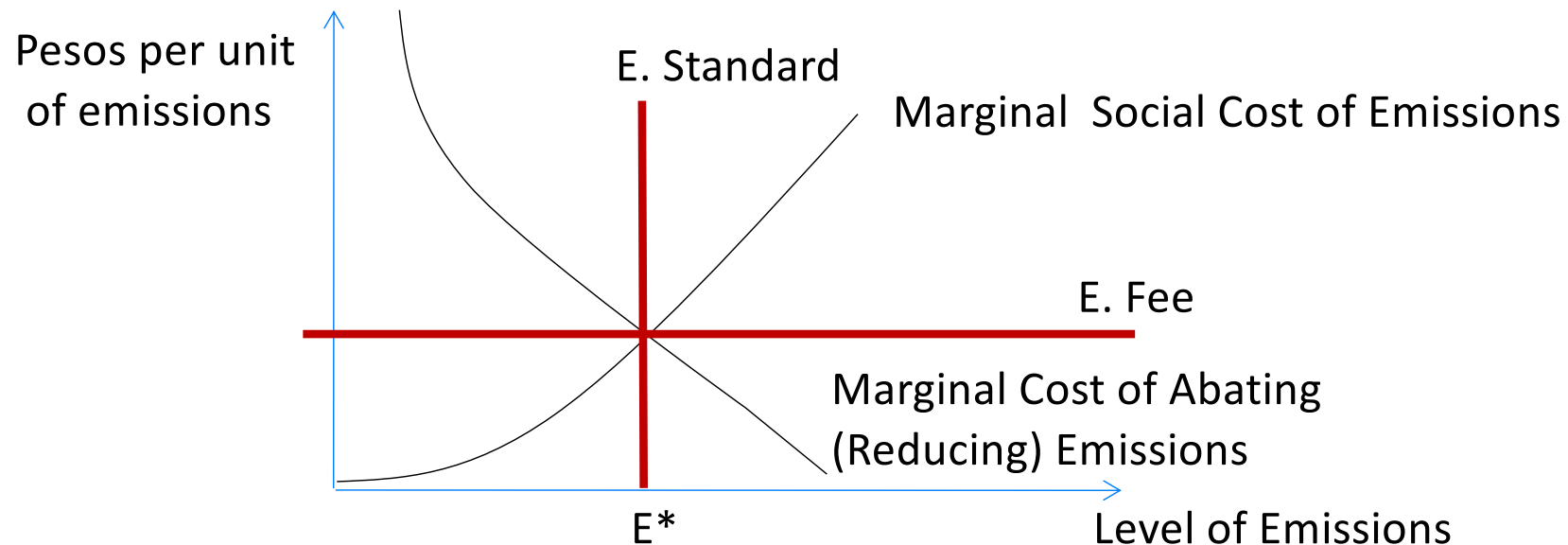
Emissions Fee

- Each firm pays for each kg of emissions produced

17. Solutions to the externality problem:

Emissions Standards and Fees

Which one is better?



Activity 1

Which one is better: **emissions standard** or **emissions fee**? Why?

17. Solutions to the externality problem: **Pollution rights**

Innovation → creation of a **market for pollution rights**

Let's imagine that x must purchase from y rights to pollute the river they share

→ x' 's choice to purchase these rights is identical to its output choice (it cannot produce without them)

Net revenue for x per unit will be: $(P_x - r)$, where r is the payment the firm must make for each unit produced

17. Solutions to the externality problem: **Pollution rights**

Firm y must decide how many rights to sell to firm x, and will be paid r for each right:

$$\pi_y = p_y g(x_i, x_o) + r x_o$$

When y maximizes profits, it must also “choose” x output! First order condition for a maximum is:

$$\partial \pi_y / \partial x_o = p_y \partial g / \partial x_o + r = 0 \rightarrow r = -p_y \partial g / \partial x_o$$

The equilibrium solution is identical to that for the Pigouvian tax

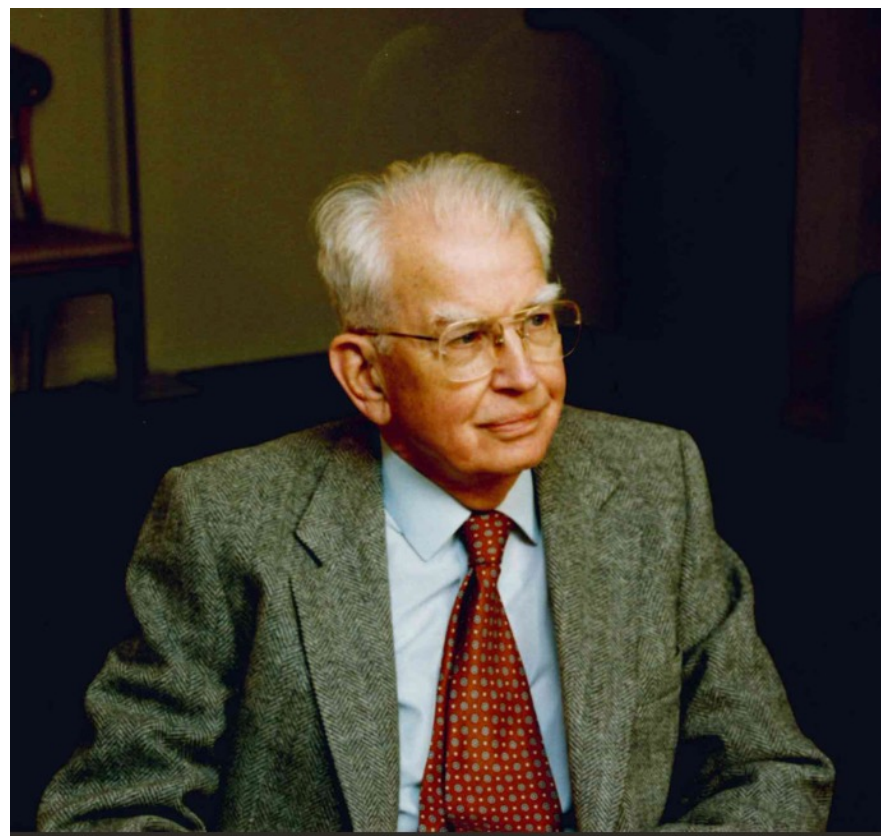
from firm x 's point of view, it makes no difference whether it pays the fee to the government or to firm y

17. Solutions to the externality problem: **Pollution rights**

The Coase Theorem → Independence of initial rights

In 1960 R. Coase showed that the key feature of the pollution rights equilibrium is that these rights be well-defined and tradable with zero transaction costs.

The initial assignment is irrelevant
→ subsequent trading will yield the same efficient equilibrium.



Ronald H. Coase, 1991. Photo: Boo Jonsson

17. Solutions to the externality problem: **Pollution rights**

The Coase Theorem → Independence of initial rights

Let's prove this → Suppose that firm x is initially given x^T rights to produce (and to pollute), it can choose to use these for its own production, or it may sell some to firm y .

Now, profits for firm x are given by

$$\pi_x = p_x x_o + r(x^T - x_o) = (p_x - r)x_o + rx^T$$

$$\pi_x = (p_x - r)f(y_i) + rx^T$$

Profits for firm y will be

$$\pi_y = p_y g(x_i, x_o) - r(x^T - x_o)$$

17. Solutions to the externality problem: **Pollution rights**

The Coase Theorem → Independence of initial rights

Although this result may seem counterintuitive, it is the assertion that all mutually beneficial transactions will be completed in the absence of impediments to making bargains.

When transaction costs are high or when information is asymmetric, initial rights assignments will matter because the sorts of trading implied by the Coase theorem may not occur.

Public Goods

Market Failures

19. Market Failures: Public Goods

Public goods are **nonexclusive** and **nonrival**

A good is **exclusive** if it is easy to exclude individuals from benefiting from the good once it is produced. A good is **nonexclusive** if it is impossible (or costly) to exclude individuals from benefiting from the good

Problem?

A good is **nonrival** if consumption of additional units of the good involves zero social marginal cost of production

Problem?

19. Market Failures: Public Goods

Public Goods: Typology of private and public goods

Some goods may possess one property but not the other:

	Exclusive	Nonexclusive
Rival	Private goods (Ice-cream, cars, houses, congested toll roads)	Common Resources (Fishing grounds, clean air, water, congested non-toll highways)
Nonrival	Club goods (mass private goods, natural monopolies) (Cable TV, tolling highways without congestion, bridges, swimming pools)	Public goods (National defense, mosquito control, justice, street lights, tsunami siren, uncongested non-toll road)

Often governments provide these goods (excepting private goods), especially for nonexclusive goods

Often nonrival goods are privately produced (swimming pools) as long as nonpayers can be excluded → Club goods

Activity 2

Are lighthouses public goods?

- a) Yes
- b) No
- c) Depends



Código de evento
QEDKEX

<https://app.wooclap.com/QEDKEX?from=instruction-slide>

Public Goods and Resource Allocation

Market Failures

20. Market Failures: Public Goods and Resource Allocation

Resource allocation problems involving public goods turn out to be quite different from resource allocation problems involving private goods.

Private markets are not a very good mechanism for allocating public goods (other social institutions –such as voting- are used)

20. Market Failures: Public Goods and Resource Allocation

Let's illustrate this mathematically:

There are 2 individuals 1 and 2 and two goods in the economy: *a public good* (B) and *x* (private good). Each individual has an income w_i , $i = 1, 2$

Both finance public good *x* by giving t_1 and t_2 , respectively. Provision of the public good is given by

$$B = t_1 + t_2$$

20. Market Failures: Public Goods and Resource Allocation

Each individual will have the following utility functions:

$$u_1(B, w_1 - t_1)$$

$$u_2(B, w_2 - t_2)$$

where

$w_i - t_i = x_i$ represents private good consumption by individual

20. Market Failures: Public Goods and Resource Allocation

Each individual will have the following utility functions:

$$u_1(\mathbf{B}, w_1 - t_1)$$

$$u_2(\mathbf{B}, w_2 - t_2)$$

where

$w_i - t_i = x_i$ represents private good consumption by individual

Activity 3

$$u_1(\mathbf{B}, w_1 - t_1) , u_2(\mathbf{B}, w_2 - t_2)$$

where $w_i - t_i = x_i$ represents private good consumption by individual i

How is nonexclusivity reflected in this model?

How is nonrivalry reflected in this model?

20. Market Failures: Public Goods and Resource Allocation

How can we find optimum t_1 and t_2 ?

$$\text{Max}_{t_1, t_2} a_1 u_1(B, w_1 - t_1) + a_2 u_2(B, w_2 - t_2)$$

$w_i - t_i = x_i$ represents private good consumption by individual

a_i represents a social weight between both individuals

20. Market Failures: Public Goods and Resource Allocation

$$\mathcal{L}(t_1, t_2) = a_1 u_1(B, w_1 - t_1) + a_2 u_2(B, w_2 - t_2)$$

Solving the first order conditions:

$$\frac{\partial \mathcal{L}}{\partial t_1} = a_1 \frac{\partial U_1}{\partial B} \cdot \frac{\partial B}{\partial t_1} + a_1 \frac{\partial U_1}{\partial x_1} \cdot \frac{\partial x_1}{\partial t_1} + a_2 \frac{\partial U_2}{\partial B} \cdot \frac{\partial B}{\partial t_1} = 0$$

$$= a_1 \frac{\partial U_1}{\partial B} \cdot 1 + a_1 \frac{\partial U_1}{\partial x_1} \cdot (-1) + a_2 \frac{\partial U_2}{\partial B} \cdot 1 = 0$$

$$a_1 \frac{\partial U_1}{\partial B} + a_2 \frac{\partial U_2}{\partial B} = a_1 \frac{\partial U_1}{\partial x_1}$$

$$\frac{\partial B}{\partial t_1} = \frac{\partial(t_1 + t_2)}{\partial t_1} = 1, \quad \frac{\partial x_1}{\partial t_1} = \frac{\partial(w_1 - t_1)}{\partial t_1} = -1$$

20. Market Failures: Public Goods and Resource Allocation

$$\mathcal{L}(t_1, t_2) = a_1 u_1(B, w_1 - t_1) + a_2 u_2(B, w_2 - t_2)$$

Similarly:

$$\frac{\partial \mathcal{L}}{\partial t_2} = a_1 \frac{\partial U_1}{\partial B} \cdot \frac{\partial B}{\partial t_2} + a_2 \frac{\partial U_2}{\partial B} \cdot \frac{\partial B}{\partial t_2} + a_2 \frac{\partial U_2}{\partial x_2} \cdot \frac{\partial x_2}{\partial t_2} = 0$$

$$= a_1 \frac{\partial U_1}{\partial B} \cdot 1 + a_2 \frac{\partial U_2}{\partial x_2} \cdot (-1) + a_2 \frac{\partial U_2}{\partial B} \cdot 1 = 0$$

$$a_1 \frac{\partial U_1}{\partial B} + a_2 \frac{\partial U_2}{\partial B} = a_2 \frac{\partial U_2}{\partial x_2}$$

$$\rightarrow a_1 \frac{\partial U_1}{\partial x_1} = a_2 \frac{\partial U_2}{\partial x_2}$$

20. Market Failures: Public Goods and Resource Allocation

Since

$$MRS_1 = \frac{\partial U_1 / \partial B}{\partial U_1 / \partial x_1} \text{ and } MRS_2 = \frac{\partial U_2 / \partial B}{\partial U_2 / \partial x_2}$$

Using one of the first order conditions:

$$a_1 \frac{\partial U_1}{\partial B} + a_2 \frac{\partial U_2}{\partial B} = a_1 \frac{\partial U_1}{\partial x_1} \quad / \cdot \frac{1}{a_1 \frac{\partial U_1}{\partial x_1}}$$
$$\frac{a_1 \frac{\partial U_1}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} + \frac{a_2 \frac{\partial U_2}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} = 1$$

20. Market Failures: Public Goods and Resource Allocation

$$\frac{a_1 \frac{\partial U_1}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} + \frac{a_2 \frac{\partial U_2}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} = 1$$

Since $a_1 \frac{\partial U_1}{\partial x_1} = a_2 \frac{\partial U_2}{\partial x_2}$

$$\frac{a_1 \frac{\partial U_1}{\partial B}}{a_1 \frac{\partial U_1}{\partial x_1}} + \frac{a_2 \frac{\partial U_2}{\partial B}}{a_2 \frac{\partial U_2}{\partial x_2}} = 1 \rightarrow MRS_1 + MRS_2 = 1$$

20. Market Failures: Public Goods and Resource Allocation

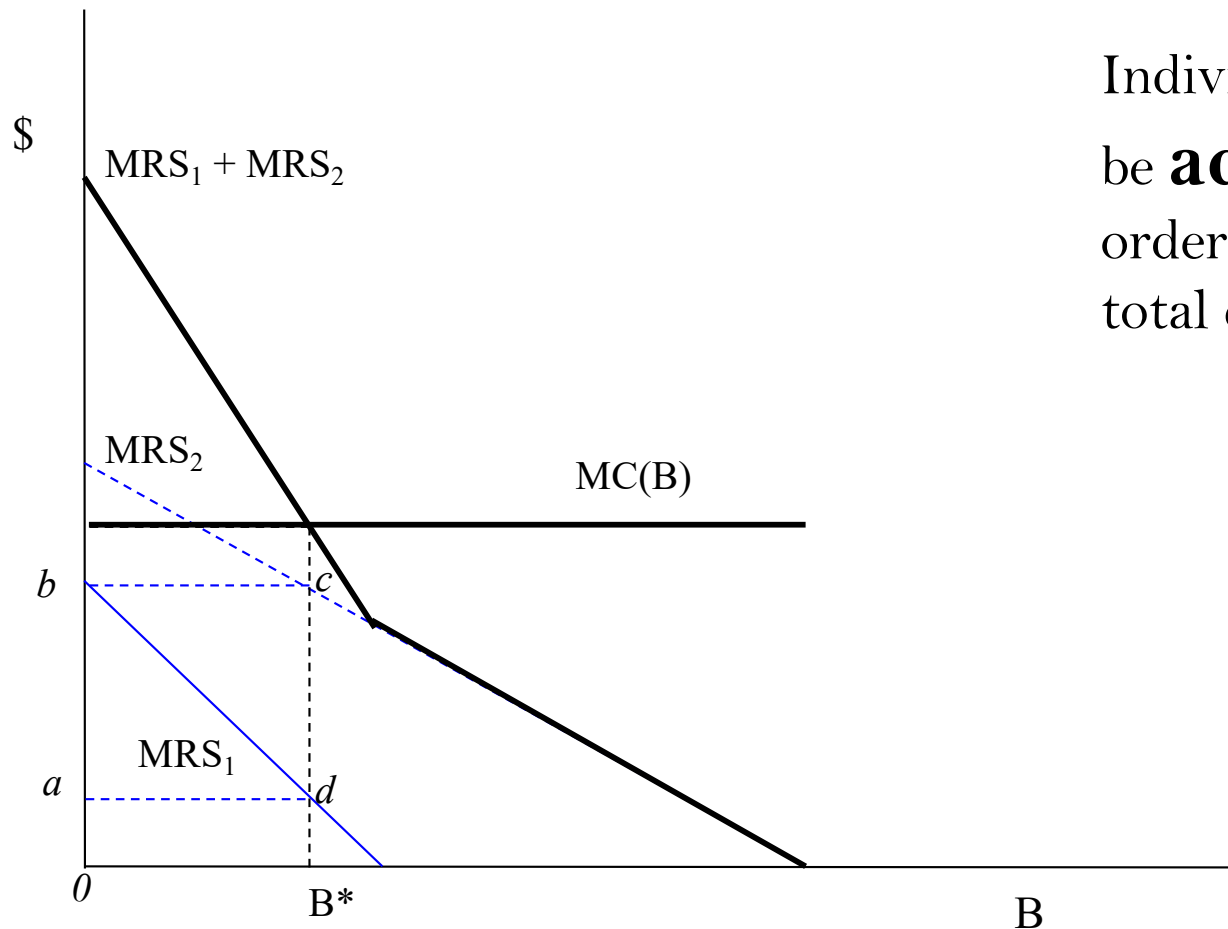
It is important to clarify that in this simple case, the **marginal cost** of providing public good B is 1.

Thus, the more general equation is that

$$MRS_1 + MRS_2 = MC(B)$$

The condition for efficiency in the case of continuous provision of the public good is that the sum of the marginal willingness-to-pay equals the marginal cost of provision

20. Market Failures: Public Goods and Resource Allocation

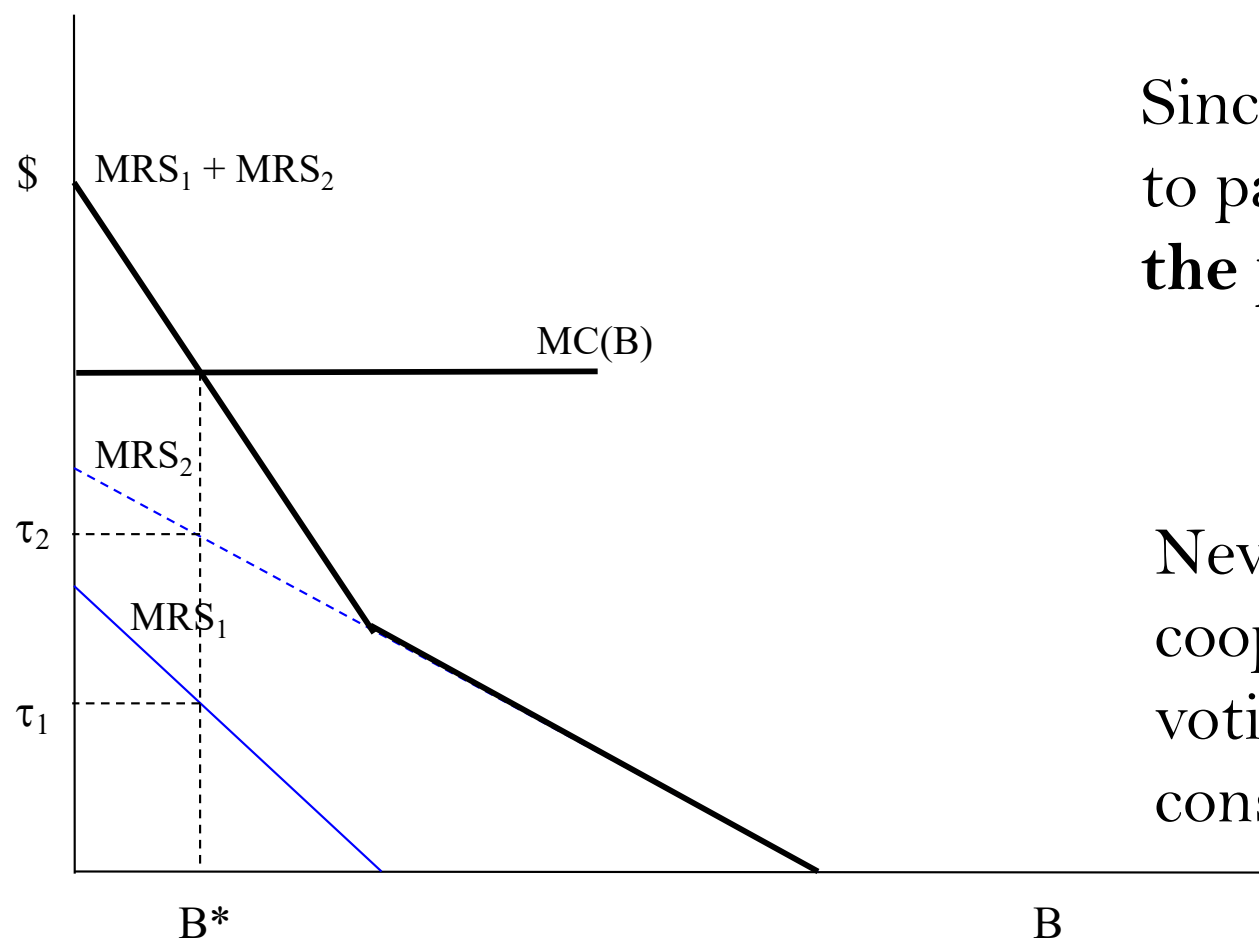


Individual demand curves must be **added vertically** in order to obtain public good's total demand.

Optimum provision quantity is B^* . Each individual should contribute to the financing of the public good the amount reflected in her demand curve for a B^* level of consumption

$$0adB^* = t_1; 0bcB^* = t_2$$

20. Market Failures: Public Goods and Resource Allocation



Since no one can be made to pay, **total provision of the public good is zero.**

Nevertheless, under a cooperation scheme or voting, the optimum level of consumption will be B^* .

20. Market Failures: Public Goods and Resource Allocation

Due to the nonrivalry and nonexclusivity, an individual knows that if the good is available, he can use it (without paying for it)

Thus, there is an incentive to reveal a lower willingness-to-pay than the real one → **free rider problem**

The market solution is an **insufficient provision of public goods**.

This is a market failure and we have to look up for alternative mechanisms of provision of public goods (voting systems, public provision → social evaluation of projects)

Efficient allocation
of public goods



Free riders
problem



Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.